

Directing Robot Motions with Paralinguistic Information

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Abstract—We propose an interface system that can extract a user’s ambiguous nuances and feelings from paralinguistic information in their expressed speech and reflect these extracted nuances in a robot’s punching action. From the result of an evaluation study, participants reported that this system succeeded in reflecting their nuances and feeling from their expressed paralinguistic information and that this system was easy and fun to use.

Keywords—component; paralinguistic information; directing robot motion; nuances and feelings.

I. INTRODUCTION

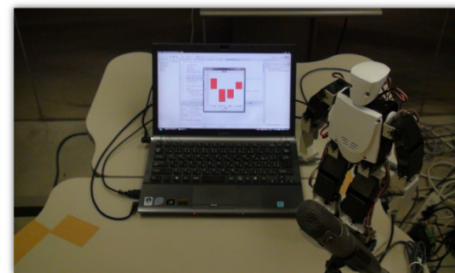
Human communications are explicitly achieved through verbal utterances as digital information, but paralinguistic information as analogue information, such as pitch, power, and the tone of utterances, also plays an important role [1]. This is because one’s internal state, such as their emotion, or ambiguous nuances in feelings are deeply reflected in their expressed paralinguistic information, which cannot be described verbally well [2]. While most studies on paralinguistic information have focused on estimating people’s emotional states from their speech [3], some studies have used paralinguistic information in order to support users in interacting with computers and robots [4,5]. However, these studies regarded certain patterns of paralinguistic information as one symbol (e.g., any increasing pitch was the symbol “increasing”), so the analogue aspects of paralinguistic information have not really been used so far.

Therefore, we propose an interface system that can extract a user’s ambiguous nuances and feelings as analogue values from paralinguistic information in their expressed speech and reflect any of these extracted nuances on a certain target that they want. In this study, we focused on robot motions as the target to which users will add their nuances and feelings. Specifically, the characteristic features of the expressed paralinguistic information were associated with the four-dimensional vector {sharpness, softness, dynamics, largeness}[6], which can represent the impressions of pure “sounds” in speech. Next, each attribute of this vector was associated with each physical feature of robot motions. Therefore, this interface system can reflect a user’s ambiguous nuances and feelings in robot motions intuitively, so this system would strongly help novice users who want to direct robot motions but do not have enough knowledge about robotics.

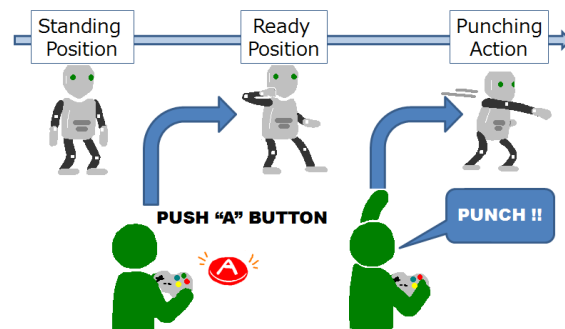
II. PROTOTYPE SYSTEM

We developed a prototype system in which nuances and feelings extracted from expressed paralinguistic information were reflected in the punching action of a humanoid robot. This system comprises a humanoid robot (Robovie-X, Vstone Co., Ltd.), a game controller (C8G-00003 for Xbox 360, Microsoft Corporation), a microphone (ECM-MS957, Sony Corporation) and a laptop PC (PCG-6X2N, Sony Corporation) with a sound analyzing module coded in Java (Figure 1(a)).

In this system, when a user presses the “A” button on the game controller, the robot changes its posture from a standing position to a ready position for punching. This robot then becomes ready to receive speech from the user. Afterward, when the user says, “punch!” to the robot via the microphone, the robot performs the punching action while reflecting their nuances extracted from the paralinguistic information in their utterance of “punch!” (Figure 1(b)).



(a) A robot, a laptop, and microphone



(b) System procedure

Figure 1. Overview of this system

III. ASSOCIATION BETWEEN PARALINGUISTIC INFORMATION, FOUR-DIMENSIONAL VECTOR, AND ROBOT MOTION

To extract the nuances and feelings from the paralinguistic information, we determined the association between the paralinguistic information, four-dimensional vector, and robot motions. First, we determined the association between the characteristic features of paralinguistic information and the four-dimensional vector $\{sharpness, softness, dynamics, largeness\}$ by means of our experimental investigation as follows.

$$\text{Sharpness} = \frac{0.8 \cdot (1 - nTime) + 0.8 \cdot nVol + nF0 + 0.8 \cdot nNum + 0.4 \cdot fType}{3.8} \quad (1)$$

$$\text{Softness} = \frac{0.2 \cdot nTime + 0.2 \cdot nVol + 0.2 \cdot (1 - nF0) + 0.8 \cdot (1 - nNum) + vType + fType}{5.4} \quad (2)$$

$$\text{Dynamic} = \frac{0.8 \cdot (1 - nTime) + 0.6 \cdot nVol + 0.8 \cdot nF0 + 0.8 \cdot nNum + 0.2 \cdot vType + 0.6 \cdot fType}{4.6} \quad (3)$$

$$\text{Largeness} = \frac{0.8 \cdot nTime + 0.8 \cdot nVol + 0.6 \cdot (1 - nF0) + 0.8 \cdot (1 - nNum) + 0.2 \cdot vType + 0.4 \cdot fType}{5.4} \quad (4)$$

Here, $nTime$, $nVol$, $nF0$, $nNum$, $vType$, and $fType$ are the average duration of input speech, average power, average fundamental frequency (F0), number of sounds, time variation of power (-1: decreasing, 0: flat, and +1: increasing), and the time variation of F0, respectively.

Next, we determined the association between each attribute of the four-dimensional vector and each physical feature of robot motions by means of the authors' consensual decision making as follows.

$$\text{Duration [msec]} = 500 + 2500 \cdot (1 - \text{Sharpness})^2 \quad (5)$$

$$\text{Acceleration} \left[\frac{\text{deg}}{\text{sec}^2} \right] = 30 - 10 \cdot \text{Softness} \quad (6)$$

$$\text{Wait [msec]} = 1500 \cdot (1 - \text{Dynamic}) \quad (7)$$

$$\text{Elbow [deg]} = 83 + 28 \cdot \text{Largeness} \quad (8-1)$$

$$\text{Shoulder [deg]} = 110 + 54 \cdot \text{Largeness} \quad (8-2)$$

Here, *Duration*, *Acceleration*, *Wait*, *Elbow*, and *Shoulder* are the duration of the punching action, acceleration of the action, wait duration until starting the punching action from the ready position, the joint angle of the elbow, and that of the shoulder, respectively. Concretely, these equations can be described as:

- A higher “sharpness” value makes the robot’s motion faster, while a lower value makes it slower.
- A higher “softness” value makes the motion’s acceleration smaller, while a lower value makes it larger.
- A higher “dynamic” value makes the wait duration of the ready position longer, while a lower value makes it shorter.
- A higher “largeness” value makes the motion of the elbow and shoulder angle larger, while a lower value makes it smaller.

IV. EVALUATION EXPERIMENT

We conducted a simple evaluation experiment to investigate whether this system can reflect a user’s ambiguous nuances and feelings from paralinguistic information when they say “punch!” to the robot. Five Japanese undergrads (4 men and 1 woman; 21 to 24 years old) participated. First, the

participants were asked to try to realize several given motions such as “strong punching” and “slowest punching” (1st phase), and they were then asked to freely use this system (2nd phase). In each phase, they were asked to evaluate whether this system could realize their intended nuance and feeling in the robot’s motions on a 7-point Likert scale (7 point maximum evaluation and 1 point minimum). Moreover, they were also asked to evaluate whether the attribute values of the four-dimensional vectors were appropriate or not on a 7-point Likert scale in order to check whether the attributes of the four-dimensional vectors were properly associated with the physical attributes of robot motions. Finally, they were asked to answer an additional four questions in order to get their subjective impressions of this system on a 7-point Likert scale: *Q1. Is this system easy to use?*, *Q2. Is it fun to use?*, *Q3. Do you want to use this at home?*, and *Q4. Do you think this system will be common?*

V. RESULTS AND FUTURE WORKS

The results are depicted in Figure 2. From this result, we could confirm that this system succeeded in reflecting the participants’ nuances and feelings extracted from their expressed paralinguistic information. The average evaluations of the additional questions were 6.4, 6.2, 5.4, and 5.6, respectively, so the participants reported that this system was easy and fun to use.

We are now planning to investigate whether this quantified paralinguistic information as four-dimensional vectors can be applied to other targets, such as drawing or music composing and arranging with computers. We believe that this study will lead to a new methodology for augmenting user creativity in various ways, and it might contribute to creating a new interface system that can reduce a user’s cognitive load when interacting with computers.

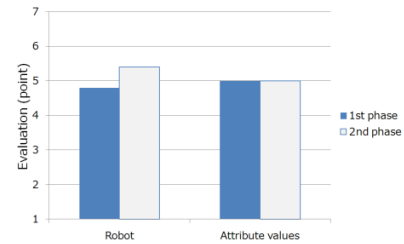


Figure 2. Results of evaluation experiment

REFERENCES

- [1] Kendon, A. Do gestures communicate? A Review. *Research on Language and Social Interaction* 27, 3 (1994), 175-200.
- [2] Cohen, P. R., Morgen, J., and Pollack, M. E. *Intentions in Communication*, The MIT Press, MA, USA, 1990.
- [3] Pierre-Yves, O. The production and recognition of emotions in speech: features and algorithms. *INT J Human-Computer St* 59, 1-2 (2003), 157-183.
- [4] Igarashi, T. and Hughes, J. F. Voice as sound: using non-verbal voice input for interactive control, In *Proc. UIST 2001*, ACM Press (2001), 155-156.
- [5] Harada, S., Wobbrock, J. O., Malkin, J., Bilmes, J. A., and Landay, J. A. Longitudinal study of people learning to use continuous voice-based cursor control, In *Proc. CHI 2009*, ACM Press (2009), 347-356.
- [6] Komatsu, T. Quantifying Japanese Onomatopoeias: Toward Augmenting Creative Activities with Onomatopoeias, In *Proc. AH12*, article No. a-15, DOI: 10.1145/2160125.2160140.